INTEGRAL CALCULUS

Indefinite Integration

1. Let $f(x)=\int x^3\sqrt{3-x^2}\,dx$. If $5f\left(\sqrt{2}\right)=-4$, then f(1) is equal to :

माना $f(x)=\int x^3\sqrt{3-x^2}\,dx$ है। यदि $5f\left(\sqrt{2}\right)=-4$ है, तो f(1) बराबर है :

(A) $-\frac{2\sqrt{2}}{5}$ (B) $-\frac{6\sqrt{2}}{5}$ (C) $-\frac{8\sqrt{2}}{5}$ (D) $-\frac{4\sqrt{2}}{5}$

Ans. B

Sol. $3-x^2=t^2$

$xdx=-t\,dt$

$=\int\left(3-t^2\right)t(-t)\,dt$

$=\int\left(t^4-3t^2\right)dt=\frac{t^5}{5}-t^3+C$

$f(x)=\frac{\left(3-x^2\right)^{5/2}}{5}-\left(3-x^2\right)^{3/2}+C$

$f\left(\sqrt{2}\right)=-\frac{4}{5}\qquad\Rightarrow C=0$

$f(1)=-\frac{6\sqrt{2}}{5}$

2. If $\int e^x\left(\frac{x\sin^{-1}x}{\sqrt{1-x^2}}+\frac{\sin^{-1}x}{\left(1-x^2\right)^{3/2}}+\frac{x}{1-x^2}\right)dx=g(x)+C$, where C is the constant of integration, then $g\left(\frac{1}{2}\right)$ equals :

यदि $\int e^x\left(\frac{x\sin^{-1}x}{\sqrt{1-x^2}}+\frac{\sin^{-1}x}{\left(1-x^2\right)^{3/2}}+\frac{x}{1-x^2}\right)dx=g(x)+C$ है, जहाँ C समाकलन–अचर है, तो $g\left(\frac{1}{2}\right)$ बराबर है :

(A) $\frac{\pi}{6}\sqrt{\frac{e}{2}}$ (B) $\frac{\pi}{6}\sqrt{\frac{e}{3}}$ (C) $\frac{\pi}{4}\sqrt{\frac{e}{2}}$ (D) $\frac{\pi}{4}\sqrt{\frac{e}{3}}$

Ans. B

Sol. $\int e^x \left(\underbrace{\frac{x\sin^{-1}x}{\sqrt{1-x^2}}}_{f(x)} + \underbrace{\frac{\sin^{-1}x}{(1-x^2)^{\frac{3}{2}}} + \frac{x}{1-x^2}}_{f^1(x)} \right) dx = g(x) + C$

$\int e^x \left(f(x) + f'(x) dx = e^x f(x) + C \right)$

So $g(x) = e^x f(x)$

$g(x) = e^x \frac{x\sin^{-1}x}{\sqrt{1-x^2}}$

$g\left(\frac{1}{2}\right) = e^{\frac{1}{2}} \times \frac{1}{2} \frac{\sin^{-1}\left(\frac{1}{2}\right)}{\sqrt{1-\frac{1}{4}}} = \frac{\pi}{6}\sqrt{\frac{e}{3}}$

3. Let $I(x) = \int \frac{dx}{(x-11)^{\frac{11}{13}}(x+15)^{\frac{15}{13}}}$. If $I(37) - I(24) = \frac{1}{4}\left(\frac{1}{b^{\frac{1}{13}}} - \frac{1}{c^{\frac{1}{13}}} \right), b, c \in N$, then $3(b+c)$ is equal to :

माना कि $I(x) = \int \frac{dx}{(x-11)^{\frac{11}{13}}(x+15)^{\frac{15}{13}}}$ है। यदि $I(37) - I(24) = \frac{1}{4}\left(\frac{1}{b^{\frac{1}{13}}} - \frac{1}{c^{\frac{1}{13}}} \right), b, c \in N$ है, तो $3(b+c)$ बराबर है :

(A) 40 (B) 39 (C) 26 (D) 22

Ans. B

Sol. $I(x) = \int \frac{dx}{(x-11)^{\frac{11}{13}}(x+15)^{\frac{15}{13}}}$

$= \int \frac{dx}{\left(\frac{x-11}{x+15}\right)^{\frac{11}{13}}(x+15)^2}$ let $\frac{x-11}{x+15} = t$

$= \frac{1}{26}\int \frac{dx}{t^{\frac{11}{13}}}$ $\frac{26dx}{(x+15)^2} = dt$

$$= \frac{1}{26}\left(\frac{t^{-\frac{11}{13}+1}}{-\frac{11}{13}+1}\right) = \frac{1}{4}t^{\frac{2}{13}} = \frac{1}{4}\left(\frac{x-11}{x+15}\right)^{\frac{2}{13}}$$

$$I(37) - I(24) = \frac{1}{4}\left\{\left(\frac{26}{52}\right)^{\frac{2}{13}} - \left(\frac{13}{39}\right)^{\frac{2}{13}}\right\}$$

$$= \frac{1}{4}\left\{\frac{1}{4^{\frac{1}{13}}} - \frac{1}{9^{\frac{1}{13}}}\right\}$$

$b = 4, c = 9$

$3(b + c) = 3(4 + 9) = 39$

4. Let $\int x^3 \sin x dx = g(x) + C$, where C is the constant of integration. If $8\left(g\left(\frac{\pi}{2}\right) + g'\left(\frac{\pi}{2}\right)\right) = \alpha\pi^3 + \beta\pi^2 + \gamma$, α, β, $\gamma \in Z$, then $\alpha + \beta - \gamma$ equals :

माना कि $\int x^3 \sin x dx = g(x) + C$ है, जहाँ C समाकलन–अचर है। यदि $8\left(g\left(\frac{\pi}{2}\right) + g'\left(\frac{\pi}{2}\right)\right) = \alpha\pi^3 + \beta\pi^2 + \gamma$, $\alpha, \beta, \gamma \in Z$ है, तो $\alpha + \beta - \gamma$ बराबर है :

(A) 48 (B) 55 (C) 62 (D) 47

Ans. B

Sol. $\int x^3 \sin x dx = g(x) + C$

x^3	+	$\sin x$
$3x^2$	–	$-\cos x$
$6x$	+	$-\sin x$
6	–	$\cos x$
0		$\sin x$

$\therefore g(x) = -x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x + C$

$$g\left(\frac{\pi}{2}\right) = \frac{3\pi^2}{4} - 6$$

$$\& \; g'\left(\frac{\pi}{2}\right) = \frac{\pi^3}{8}$$

$$8\left[g\left(\frac{\pi}{2}\right)+g'\left(\frac{\pi}{2}\right)\right]=\pi^3+6\pi^2-48$$

$\alpha+\beta-\gamma=55$

5. If $f(x)=\int\frac{1}{x^{\frac{1}{4}}\left(1+x^{\frac{1}{4}}\right)}dx$, f(0) = –6, then f(1) is equal to :

यदि $f(x)=\int\frac{1}{x^{\frac{1}{4}}\left(1+x^{\frac{1}{4}}\right)}dx$, f(0) = –6, है, तो f(1) बराबर है :

(A) $2-\log_e 2$ (B) $4(\log_e 2-2)$ (C) $\log_e 2+2$ (D) $4(\log_e 2+2)$

Ans. B

Sol. $f(x)=\int\frac{1}{x^{\frac{1}{4}}\left(1+x^{\frac{1}{4}}\right)}dx$

$x^{\frac{1}{4}}=t$

$\left(\frac{1}{4}\right)x^{\frac{-3}{4}}dx=dt$

$f(x)=\int\frac{4\cdot x^{\frac{3}{4}}dt}{t(1+t)}$

$\Rightarrow\int\frac{4t^3}{t(1+t)}dt$

$f(x)=4\int\frac{(t^2-1)+1}{(1+t)}dt$

$f(x)=4\int\left[(t-1)+\frac{1}{(1+t)}\right]dt$

$f(x)=4\left(\frac{t^2}{2}-t+\ell n(1+t)\right)+C$

$$f(x)=4\left(\frac{x^{\frac{1}{2}}}{2}-x^{\frac{1}{4}}+\ell n\left(1+x^{\frac{1}{4}}\right)\right)+C$$

If $f(0)=-6$

$f(0)=4(\ell n(1))+C=-6$

$C=-6$

$$f(x)=4\left(\frac{x^{\frac{1}{2}}}{2}-x^{\frac{1}{4}}+\ell n\left(1+x^{\frac{1}{4}}\right)\right)-6$$

$$f(1)=4\left(\frac{1}{2}-1+\ell n2\right)-6$$

$$\Rightarrow 4\left(\ell n2-\frac{1}{2}\right)-6$$

$\Rightarrow 4\ell n2-2-6$

$\Rightarrow (4\ell n2-8)\Rightarrow 4(\ell n2-2)$

6. If $\int\frac{1}{a^2\sin^2 x+b^2\cos^2 x}dx=\frac{1}{12}\tan^{-1}(3\tan x)+$ constant, then the maximum value of $a\sin x+b\cos x$, is :

यदि $\int\frac{1}{a^2\sin^2 x+b^2\cos^2 x}dx=\frac{1}{12}\tan^{-1}(3\tan x)+$ अचर है, तो $\sin x+b\cos x$ का उच्चतम मान है :

(A) $\sqrt{40}$ (B) $\sqrt{42}$ (C) $\sqrt{39}$ (D) $\sqrt{41}$

Ans. A

Sol. $\int\frac{\sec^2 x dx}{a^2\tan^2 x+b^2}$

let $\tan x=t$

$\sec^2 dx=dt$

$$\int\frac{dt}{a^2t^2+b^2}$$

$$\frac{1}{a^2}\int\frac{dt}{t^2+\left(\frac{b}{a}\right)^2}$$

$$\frac{1}{a^2}\frac{1}{\frac{b}{a}}\tan^{-1}\left(\frac{t}{b}a\right)+c$$

$$\frac{1}{ab}\tan^{-1}\left(\frac{\alpha}{b}\tan x\right)+c$$

on compairing $\frac{a}{b}=3$

ab = 12

a = 6, b = 2

maximum value of

6 sinx + 2cosx is $\sqrt{40}$

7. Let $I(x)=\int\frac{6}{\sin^2 x(1-\cot x)^2}dx$. If I(0) = 3, then $I\left(\frac{\pi}{12}\right)$ is equal to :

माना $I(x)=\int\frac{6}{\sin^2 x(1-\cot x)^2}dx$ है। यदि I(0) = 3, तब $I\left(\frac{\pi}{12}\right)$ बराबर होगा :

(A) $2\sqrt{3}$ (B) $\sqrt{3}$ (C) $3\sqrt{3}$ (D) $6\sqrt{3}$

Ans. C

Sol. $I(x)=\int\frac{6dx}{\sin^2(1-\cot x)^2}$ $I(0)=3$

$I\left(\frac{\pi}{12}\right)=?$

$$I(x)=\int\frac{6dx}{\sin^2 x\frac{(\sin x-\cos x)^2}{\sin^2 x}}=\int\frac{6dx}{\sin^2 x+\cos^2 x-2\sin\cos x}$$

$$=\int\frac{6dx}{1-2\sin x\cos x}=\int\frac{6\sec^2 xdx}{1+\tan^2 x-2\tan x}$$

$$=\int\frac{6\sec^2 xdx}{(\tan x-1)^2}$$

= tan x = t

= $\sec^2 x\,dx = dt$

$$= \int \frac{6dt}{(t-1)^2} = 6\int (t-1)^{-2}\, dt$$

$$I(x) = \frac{-6}{(t-1)} + C$$

$$I(x) = -\frac{6}{\tan x - 1} + C$$

$I(0) = 3$

$3 = 6 + C$

$C = -3$

$$I(x) = -\frac{6}{\tan x - 1} - 3$$

$$I\left(\frac{\pi}{12}\right) = -\frac{6}{2-\sqrt{3}-1} - 3$$

$$= -\frac{6}{1-\sqrt{3}} - 3 = -\left(\frac{6+3-3\sqrt{3}}{1-\sqrt{3}}\right)$$

$$= -\left(\frac{9-3\sqrt{3}}{1-\sqrt{3}} \times \frac{1+\sqrt{3}}{1+\sqrt{3}}\right)$$

$$= -\left(\frac{9+9\sqrt{3}-3\sqrt{3}-9}{1-3}\right) = \frac{6\sqrt{3}}{2} = 3\sqrt{3}$$

8. Let $\int \frac{2-\tan x}{3+\tan x} dx = \frac{1}{2}\left(\alpha x + \log_e |\beta \sin x + \gamma \cos x|\right) + C$, where C is the constant of integration. Then $\alpha + \frac{\gamma}{\beta}$ is equal to :

माना $\int \frac{2-\tan x}{3+\tan x} dx = \frac{1}{2}\left(\alpha x + \log_e |\beta \sin x + \gamma \cos x|\right) + C$ है, जहाँ C समाकलन अचर है। तो $\alpha + \frac{\gamma}{\beta}$ बराबर है :

(A) 3 (B) 1 (C) 7 (D) 7

Ans. D

Sol. $\int \frac{2-\tan x}{3+\tan x} dx = \int \frac{2\cos x - \sin x}{3\cos x + \sin x} dx$

Put, $2\cos x - \sin x = a(-3\sin x + \cos x) + b(3\cos x + \sin x)$

$a + 3b = 2$ .....(i)

$-3a + b = -1$ .....(ii)

From equation (i) and (ii) $a = b = \frac{1}{2}$

$$I = \frac{1}{2}\int \frac{-3\sin x + \cos x}{3\cos x + \sin x}dx + \frac{1}{2}\int \frac{3\cos x + \sin x}{3\cos x + \sin x}dx$$

$$I = \frac{1}{2}\ell n\left|3\cos x + \sin x\right| + \frac{1}{2}x = \frac{1}{2}\left(\alpha x + \log\left|\beta\sin x + \gamma\cos x\right|\right)$$

On comparing, we get

$\alpha = 1, \beta = 1, \gamma = 3$

$$\alpha + \frac{\gamma}{\beta} = 1 + \frac{3}{1} = 4$$

9. The integral $\int \frac{\left(x^8 - x^2\right)dx}{\left(x^{12} + 3x^6 + 1\right)\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)}$ is equal to :

समाकलन $\int \frac{\left(x^8 - x^2\right)dx}{\left(x^{12} + 3x^6 + 1\right)\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)}$ बराबर है :

(A) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{1/3} + C$

(B) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{1/2} + C$

(C) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right) + C$

(D) $\log_e\left(\left|\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)\right|\right)^{3} + C$

Ans. A

Sol. $\int \frac{\left(x^8 - x^2\right)dx}{\left(x^{12} + 3x^6 + 1\right)\tan^{-1}\left(x^3 + \frac{1}{x^3}\right)}$

$$\tan^{-1}\left(x^3 + \frac{1}{x^3}\right) = t$$

$$\frac{\left(3x^2 - \frac{3}{x^4}\right)dx}{1 + \left(x^3 + \frac{1}{x^3}\right)^2} = dt$$

$$\frac{3\left(x^6-1\right)dx}{x^4\left(1+x^6+\frac{1}{x^6}+2\right)}=dt$$

$$\frac{3\left(x^8-x^2\right)dx}{\left(x^{12}+3x^6+1\right)}=dt$$

$$=\frac{1}{3}\int\frac{dt}{t}$$

$$=\frac{1}{3}\ell n|t|+c$$

$$\log_e\left|\tan^{-1}\left(x^3+\frac{1}{x^3}\right)\right|^{1/3}+c$$

10. For $x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, if $y(x)=\int\frac{\cos ec\,x+\sin x}{\cos ecx\sec x+\tan x\sin^2 x}dx$, and $\lim\limits_{x\to\left(\frac{\pi}{2}\right)^-}y(x)=0$ then $y\left(\frac{\pi}{4}\right)$ is equal to :

$x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ के लिए यदि $y(x)=\int\frac{\cos ec\,x+\sin x}{\cos ecx\sec x+\tan x\sin^2 x}dx$ है तथा $\lim\limits_{x\to\left(\frac{\pi}{2}\right)^-}y(x)=0$ है, तो $y\left(\frac{\pi}{4}\right)$ बराबर है :

(A) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (B) $\frac{1}{\sqrt{2}}\tan^{-1}\left(-\frac{1}{2}\right)$ (C) $-\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (D) $\frac{1}{2}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

Ans. B

Sol. $\int\left(\frac{1+\sin^2 x}{\frac{1}{\cos x}+\frac{\sin^4 x}{\cos x}}\right)dx$

$$=\int\frac{\left(1+\sin^2 x\right)\cos x}{1+\sin^4 x}dx$$

$\sin x=t,\qquad \cos x\,dx=dt$

$$\int \frac{1+t^2}{1+t^4}dt = \int \left(\frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}} \right) dt$$

$$= \int \frac{du}{u^2+2} \qquad t-\frac{1}{t}=u$$

$$= \frac{1}{\sqrt{2}} \tan^{-1} \frac{u}{\sqrt{2}} + c \qquad \left(1+\frac{1}{t^2}\right)dt = dx$$

$$f(x) = \frac{1}{\sqrt{2}} \tan^{-1} \frac{\left(\sin x - \frac{1}{\sin x}\right)}{\sqrt{2}} + c$$

$$f\left(\frac{\pi}{2}\right) = 0 \Rightarrow c = 0$$

$$f\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\frac{1}{\sqrt{2}} - \sqrt{2}}{\sqrt{2}} \right)$$

$$f\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \tan^{-1} \left(-\frac{1}{2}\right)$$

11. If $\int \frac{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x}{\sqrt{\sin^3 x \cos^3 x \sin(x-\theta)}} dx = A\sqrt{\cos\theta \tan x - \sin\theta} + B\sqrt{\cos\theta - \sin\theta \cot x} + C$, where C is the integration constant, then AB is equal to :

यदि $\int \frac{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x}{\sqrt{\sin^3 x \cos^3 x \sin(x-\theta)}} dx = A\sqrt{\cos\theta \tan x - \sin\theta} + B\sqrt{\cos\theta - \sin\theta \cot x} + C$ है, जहां C समाकलन अचर है, तो AB बराबर है :

(A) 4 sec θ (B) 4 cosec (2θ) (C) 2 sec θ (D) 8 cosec (2θ)

Ans. D

Sol. $I=\int\frac{\left(\sin^{\frac{3}{2}}x+\cos^{\frac{3}{2}}x\right)dx}{\sqrt{\sin^3 x\cos^3 x\sin(x-\theta)}}$

$$I=\int\frac{\sec^{\frac{3}{2}}x dx}{\sqrt{\sin(x-\theta)}}+\int\frac{\text{cosec}^{\frac{3}{2}}x dx}{\sqrt{\sin(x-\theta)}}$$

$$I=\int\frac{\sec^2 x dx}{\sqrt{\sec x\sin(x-\theta)}}+\int\frac{\text{cosec}^2 x dx}{\sqrt{\text{cosec}\, x\sin(x-\theta)}}$$

$$I=\int\frac{\sec^2 x dx}{\sqrt{\tan x\cos\theta-\sin\theta}}+\int\frac{\text{cosec}^2 x dx}{\sqrt{\cos\theta-\cot x\sin\theta}}=I_1+I_2$$

$$I_1=\int\frac{\sec^2 x dx}{\sqrt{\tan x\cos\theta-\sin\theta}}$$

Let $\tan x\cos\theta-\sin\theta=t^2$

$$\Rightarrow \sec^2 x dx=\frac{2tdt}{\cos\theta}$$

$$I_1=\int\frac{\frac{2tdt}{\cos\theta}}{t}=\frac{2t}{\cos\theta}+C_1$$

$$=\frac{2}{\cos\theta}\sqrt{\tan x\cot\theta-\sin\theta}+C_1$$

$$I_2=\frac{2}{\sin\theta}\sqrt{\cos\theta-\sin\theta\cot x}+C_2$$

$$\Rightarrow I_1+I_2\Rightarrow B=\frac{2}{\sin\theta},A=\frac{2}{\cos\theta}$$

$$\Rightarrow AB=\frac{4}{\sin\theta\cos\theta}=8\,\text{cosec}\,2\theta$$

12. Let $I(x)=\int\frac{x^2\left(x\sec^2 x+\tan x\right)}{(x\tan x+1)^2}dx$. If $I(0)=0$, then $I\left(\frac{\pi}{4}\right)$ is equal to :

माना $I(x)=\int\frac{x^2\left(x\sec^2 x+\tan x\right)}{(x\tan x+1)^2}dx$ है। यदि $I(0)=0$ है, तब $I\left(\frac{\pi}{4}\right)$ का मान है :

(A) $\log_e \frac{(\pi+4)^2}{16}+\frac{\pi^2}{4(\pi+4)}$ (B) $\log_e \frac{(\pi+4)^2}{32}-\frac{\pi^2}{4(\pi+4)}$

(C) $\log_e \frac{(\pi+4)^2}{32}+\frac{\pi^2}{4(\pi+4)}$ (D) $\log_e \frac{(\pi+4)^2}{16}-\frac{\pi^2}{4(\pi+4)}$

Ans. B

Sol. $\int x^2\left(\frac{x\sec^2 x+\tan x}{(x\tan x+1)^2}\right)dx$

$=\frac{x^2}{(x\tan x+1)}+\int\frac{2x}{x\tan x+1}dx$

$I=2\int\frac{x}{x\tan x+1}dx$

$=2\int\frac{x\cos x}{x\sin x+\cos x}dx$

Let $x\sin x+\cos x=t$

$(x\cos x+\sin x-\sin x)dx=dt$

$=2\int\frac{dt}{t}=2\log t+c$

$=2\log|x\sin x+\cos x|+c$

$\therefore\int\frac{x^2\left(x\sec^2 x+\tan x\right)}{\left(x\tan+1\right)^2}dx=\frac{-x^2}{x\tan x+1}+2\log\left|x\sin x+\cos x\right|+c$

$I(0)=0 \quad\Rightarrow c=0$

$I\left(\frac{\pi}{4}\right)=\frac{-\pi}{\frac{\pi}{4}\times1+1}+2\log\left|\frac{1}{\sqrt{2}}\left(\frac{\pi}{4}\right)+1\right|$

$=\log_e\frac{(\pi+4)^2}{32}-\frac{\pi^2}{4(\pi+4)}$

13. If $I(x)=\int e^{\sin^2 x}(\cos x\sin 2x-\sin x)dx$ and $I(0)=1$, then $I\left(\frac{\pi}{3}\right)$ is equal to :

यदि $I(x)=\int e^{\sin^2 x}(\cos x\sin 2x-\sin x)dx$ तथा $I(0)=1$ हैं, तो $I\left(\frac{\pi}{3}\right)$ बराबर है :

(A) $e^{\frac{3}{4}}$ (B) $-\frac{1}{2}e^{\frac{3}{4}}$ (C) $-e^{\frac{3}{4}}$ (D) $\frac{1}{2}e^{\frac{3}{4}}$

Ans. D

Sol. $\because I=\int e^{f(x)}\left(g'(x)+f'(x)\cdot g(x)\right)dx=e^{f(x)}\cdot g(x)$

$\therefore I=e^{\sin^2 x}\cdot\cos x+c$

but $I(0)=1\Rightarrow c=0$

$I=e^{\sin^2 x}\cos x$

$I\left(\frac{\pi}{3}\right)=\frac{1}{2}e^{\frac{3}{4}}$

14. For $\alpha, \beta, \gamma, \delta\in\mathbb{N}$, if $\int\left(\left(\frac{x}{e}\right)^{2x}+\left(\frac{e}{x}\right)^{2x}\right)\log_e x\,dx=\frac{1}{\alpha}\left(\frac{x}{e}\right)^{\beta x}-\frac{1}{\gamma}\left(\frac{e}{x}\right)^{\delta x}+C$, where $e=\sum_{n=0}^{\infty}\frac{1}{n!}$ and C is constant of integration, then $\alpha+2\beta+3\gamma-4\delta$ is equal to :

$\alpha, \beta, \gamma, \delta\in\mathbb{N}$ के लिए, यदि $\int\left(\left(\frac{x}{e}\right)^{2x}+\left(\frac{e}{x}\right)^{2x}\right)\log_e x\,dx=\frac{1}{\alpha}\left(\frac{x}{e}\right)^{\beta x}-\frac{1}{\gamma}\left(\frac{e}{x}\right)^{\delta x}+C$ है, जहाँ $e=\sum_{n=0}^{\infty}\frac{1}{n!}$ तथा C समाकलन अचर है, तो $\alpha+2\beta+3\gamma-4\delta$ बराबर है :

(A) –4 (B) 1 (C) 4 (D) –8

Ans. C

Sol. Let $\left(\frac{x}{e}\right)^{2x}=t$

$2x(\ln x-1)=\ln t$

$\left(2(\ln x-1)+2x\times\frac{1}{x}\right)dx=\frac{1}{t}dt$

$\ln x\,dx=\frac{1}{2t}dt$

$I=\int\left(t+\frac{1}{t}\right)\times\frac{1}{2t}dt=\frac{1}{2}\int\left(1+\frac{1}{t^2}\right)dt$

$=\frac{1}{2}\left(t-\frac{1}{t}\right)+C$

$$=\frac{1}{2}\left(\frac{x}{e}\right)^{2x}-\frac{1}{2}\left(\frac{e}{x}\right)^{2x}+C$$

$\Rightarrow \alpha=2,\beta=2,\gamma=2,\delta=2$

$\therefore \quad \alpha+2\beta+3\gamma-4\delta=2+4+6-8=4$

15. If $\int\frac{\left(\sqrt{1+x^2}+x\right)^{10}}{\left(\sqrt{1+x^2}-x\right)^{9}}dx=\frac{1}{m}\left(\left(\sqrt{1+x^2}+x\right)^{n}\left(n\sqrt{1+x^2}-x\right)\right)+C$ where C is the constant of integration and m, $n\in N$, then $m+n$ is equal to _____.

यदि $\int\frac{\left(\sqrt{1+x^2}+x\right)^{10}}{\left(\sqrt{1+x^2}-x\right)^{9}}dx=\frac{1}{m}\left(\left(\sqrt{1+x^2}+x\right)^{n}\left(n\sqrt{1+x^2}-x\right)\right)+C$ है, जहाँ C समाकलन अचर है और $m, n\in N$ हैं, तो $m + n$ बराबर है _____.

Ans. 379

Sol. $\int\frac{\left(\sqrt{1+x^2}+x\right)^{10}}{\left(\sqrt{1+x^2}-x\right)^{9}}dx$

Put $x=\tan\theta$

$\int\frac{(\sec\theta+\tan\theta)^{10}}{(\sec\theta-\tan\theta)^{9}}\left(\sec^2\theta d\theta\right)$

$\int(\sec\theta+\tan\theta)^{19}\left(\sec^2\theta\right)d\theta$

Put $\sec\theta+\tan\theta=z$; $\sec\theta(\sec\theta+\tan\theta)d\theta=dz$

$\sec\theta-\tan\theta=\frac{1}{z}$

$\sec\theta=\frac{1}{2}\left(z+\frac{1}{z}\right)$

$\Rightarrow\int z^{18}\left(\frac{1}{2}\left(z+\frac{1}{z}\right)\right)dz$

$\Rightarrow\frac{1}{2}\int\left(z^{19}+z^{17}\right)dz$

$$\Rightarrow \frac{1}{2}\left(\frac{z^{20}}{20}+\frac{z^{18}}{18}\right)+c$$

$$=\frac{1}{2}\left(\frac{(\sec\theta+\tan\theta)^{20}}{20}+\frac{(\sec\theta+\tan\theta)^{18}}{18}\right)+c$$

$$=\frac{1}{2}\times\frac{\left(\sqrt{1+x^2}+x\right)^{18}}{2}\left\{\frac{\left(\sqrt{1+x^2}+x\right)^{2}}{10}+\frac{1}{9}\right\}+c$$

$$\Rightarrow \frac{1}{4}\left(\sqrt{1+x^2}+x\right)^{19}\left\{\frac{\sqrt{1+x^2}+x}{10}+\frac{1}{9}\left(\sqrt{1+x^2}-x\right)\right\}+c$$

$$\Rightarrow \frac{1}{360}\left(\sqrt{1+x^2}+x\right)^{19}\left\{19\left(\sqrt{1+x^2}\right)-x\right\}+c$$

$\Rightarrow m = 360, n = 19$

16. If $\int\left(\frac{1}{x}+\frac{1}{x^3}\right)\left(\sqrt[23]{3x^{-24}+x^{-26}}\right)dx=-\frac{\alpha}{3(\alpha+1)}\left(3x^{\beta}+x^{\gamma}\right)^{\frac{\alpha+1}{\alpha}}+C$, $x>0$, $(\alpha, \beta, \gamma \in Z)$, where C is the constant of integration, then $\alpha+\beta+\gamma$ is equal to _____.

यदि $\int\left(\frac{1}{x}+\frac{1}{x^3}\right)\left(\sqrt[23]{3x^{-24}+x^{-26}}\right)dx=-\frac{\alpha}{3(\alpha+1)}\left(3x^{\beta}+x^{\gamma}\right)^{\frac{\alpha+1}{\alpha}}+C$, $x>0$, $(\alpha, \beta, \gamma \in Z)$, जहाँ C समाकलन अचर है, तो $\alpha+\beta+\gamma$ बराबर है _____.

Ans. 19

Sol. $I=\int\left(\frac{1}{x}+\frac{1}{x^3}\right)\left(\frac{3}{x^{24}}+\frac{1}{x^{26}}\right)^{\frac{1}{23}}dx$

$$I=\int\left(\frac{1}{x^2}+\frac{1}{x^4}\right)\left(\frac{3}{x}+\frac{1}{x^3}\right)^{\frac{1}{23}}dx$$

Let $\frac{3}{x}+\frac{1}{x^3}=t$

$$\Rightarrow\left(-\frac{3}{x^2}-\frac{3}{x^4}\right)dx=dt$$

so, $I = -\frac{1}{3}\int t^{\frac{1}{23}}dt$

$= -\frac{1}{3}\cdot\frac{23}{24}t^{\frac{24}{23}} + c$

$= -\frac{1}{3}\cdot\frac{23}{24}\left(\frac{3}{x}+\frac{1}{x^3}\right)^{\frac{24}{23}} + c$

$= -\frac{1}{3}\cdot\frac{23}{24}\left(3x^{-1}+x^{-3}\right)^{\frac{24}{23}} + c$

comparing $\alpha = 23, \beta = -1, \gamma = -3$

$\Rightarrow \alpha + \beta + \gamma = 19$

17. If $\int\frac{2x^2+5x+9}{\sqrt{x^2+x+1}}dx = x\sqrt{x^2+x+1}+\alpha\sqrt{x^2+x+1}+\beta\log_e\left|x+\frac{1}{2}+\sqrt{x^2+x+1}\right|+C$, where C is the constant of integration, then $\alpha+2\beta$ is equal to _____.

यदि $\int\frac{2x^2+5x+9}{\sqrt{x^2+x+1}}dx = x\sqrt{x^2+x+1}+\alpha\sqrt{x^2+x+1}+\beta\log_e\left|x+\frac{1}{2}+\sqrt{x^2+x+1}\right|+C$ है, जहाँ C समाकलन अचर है, तो $\alpha + 2\beta$ बराबर है _____.

Ans. 16

Sol. Diff^n both sides;

$$\frac{2x^2+5x+9}{\sqrt{x^2+x+1}} = x\left(\frac{2x+1}{2\sqrt{x^2+x+1}}\right)+\sqrt{x^2+x+1}+\alpha\left(\frac{2x+1}{2\sqrt{x^2+x+1}}\right)+\beta\left(\frac{1+\frac{2x+1}{2\sqrt{x^2+x+1}}}{x+\frac{1}{2}+\sqrt{x^2+x+1}}\right)$$

$$\Rightarrow \frac{2x^2+5x+9}{\sqrt{x^2+x+1}} = \frac{2x^2+x+2x^2+2x+2+2\alpha x+\alpha}{2\sqrt{x^2+x+1}}+\beta\left(\frac{2\sqrt{x^2+x+1}+2x+1}{2\sqrt{x^2+x+1}}\right)\left(\frac{2}{2x+1+2\sqrt{x^2+x+1}}\right)$$

$$\Rightarrow \frac{2x^2+5x+9}{\sqrt{x^2+x+1}} = \frac{4x^2+(3+2\alpha)x+2+\alpha+2\beta}{2\sqrt{x^2+x+1}}$$

$$\Rightarrow 5 = \frac{3+2\alpha}{2} \text{ and } 9 = \frac{2+\frac{7}{2}+2\beta}{2}$$

$\alpha = \frac{7}{2}$ $\quad 16 - \frac{7}{2} = 2\beta$ $\quad \Rightarrow \beta = \frac{25}{4}$

$\Rightarrow \alpha + 2\beta = \frac{7}{2} + \frac{25}{2} = 16$

18. If $\int \text{cosec}^5 x dx = \alpha \cot x \, \text{cosec} x \left(\text{cosec}^2 x + \frac{3}{2} \right) + \beta \log_e \left| \tan \frac{x}{2} \right| + C$ where $\alpha, \beta \in R$ and C is the constant of integration, then the value of $8(\alpha + \beta)$ equals _____.

यदि $\int \text{cosec}^5 x dx = \alpha \cot x \, \text{cosec} x \left(\text{cosec}^2 x + \frac{3}{2} \right) + \beta \log_e \left| \tan \frac{x}{2} \right| + C$ है, जहाँ $\alpha, \beta \in R$ तथा C समाकल अचर है, तो $8(\alpha + \beta)$ का मान बराबर है _____

Ans. 1

Sol. $\int \text{cosec}^3 x \cdot \text{cosec}^2 x dx = I$

By applying integration by parts

$I = -\cot x \, \text{cosec}^3 x + \int \cot x \left(-3\text{cosec}^2 x \cot x \text{cosec} x \right) dx$

$I = -\cot x \, \text{cosec}^3 x - 3I - 3\int \text{cosec}^3 x \left(\text{cosec}^2 x - 1 \right) dx$

$I = -\cot x \, \text{cosec}^3 x - 3I + 3\int \text{cosec}^3 x dx$

Let

$I_1 = \int \text{cosec}^3 x dx = -\text{cosec} x \cot x - \int \cot^2 x \, \text{cosec} x dx$

$I_1 = -\text{cosec} x \cot x - \int \left(\text{cosec}^2 x - 1 \right) \text{cosec} x dx$

$2I_1 = -\text{cosec} x \cot x + \ell n \left| \tan \frac{x}{2} \right|$

$I_1 = -\frac{1}{2} \text{cosec} x \cot x + \frac{1}{2} \ell n \left| \tan \frac{x}{2} \right|$

$4I = -\cot x \, \text{cosec}^3 x - \frac{3}{2} \text{cosec} x \cot x + \frac{3}{2} \ell n \left| \tan \frac{x}{2} \right| + 4c$

$I = -\frac{1}{4} \text{cosec} x \cot x \left(\text{cosec}^2 x + \frac{3}{2} \right) + \frac{3}{2} \ell n \left| \tan \frac{x}{2} \right| + c$

$\therefore \alpha = \frac{-1}{4}, \ \beta = \frac{3}{8} \rightarrow 8(\alpha + \beta) = 1$

19. Let $I(x)=\int\sqrt{\frac{x+7}{x}}dx$ and $I(9)=12+7\log_e 7$. If $I(1)=\alpha+7\log_e(1+2\sqrt{2})$, α^4 is equal to _____.

माना $I(x)=\int\sqrt{\frac{x+7}{x}}dx$ तथा $I(9)=12+7\log_e 7$ हैं। यदि $I(1)=\alpha+7\log_e(1+2\sqrt{2})$ है, तो α^4 बराबर है _____

Ans. 64

Sol. $I(x)=\int\sqrt{\frac{x+7}{x}}dx$

$$=\int\frac{\sqrt{\left(\sqrt{x}\right)^2+\left(\sqrt{7}\right)^2}}{\sqrt{x}}dx$$

Let $\sqrt{x}=t\therefore\frac{1}{2\sqrt{x}}dx=dt$

$$=2\int\sqrt{t^2+(\sqrt{7})^2}dt=2\left(\frac{t}{2}\sqrt{t^2+7}+\frac{7}{2}\ln\left(t+\sqrt{t^2+7}\right)\right)+c$$

$I(x)=\sqrt{x}\sqrt{x+7}+7\ln(\sqrt{x}+\sqrt{x+7})+c$

$I(9)=12+7\ln 7=12+7\ln(3+4)+c$

$\Rightarrow c=0$

$\therefore I(x)=\sqrt{x}\sqrt{x+7}+7\ln(\sqrt{x}+\sqrt{x+7})$

at $x=1$

$I(1)=\sqrt{8}+7\ell n(1+\sqrt{8})=7\ell n(1+2\sqrt{2})+\alpha$

$\alpha=\sqrt{8}$

$\alpha^4=64$

20. Let $f(x)=\int\frac{dx}{\left(3+4x^2\right)\sqrt{4-3x^2}}$, $|x|<\frac{2}{\sqrt{3}}$. If $f(0)=0$ and $f(1)=\frac{1}{\alpha\beta}\tan^{-1}\left(\frac{\alpha}{\beta}\right)$, $\alpha,\beta>0$, then $\alpha^2+\beta^2$ is equal to _____.

यदि $f(x)=\int\frac{dx}{\left(3+4x^2\right)\sqrt{4-3x^2}}$, $|x|<\frac{2}{\sqrt{3}}$, $f(0)=0$ तथा $f(1)=\frac{1}{\alpha\beta}\tan^{-1}\left(\frac{\alpha}{\beta}\right)$, $\alpha,\beta>0$ हो, तो $\alpha^2+\beta^2$ का मान होगा _____

Ans. 28

Sol. $f(x)=\int\frac{dx}{\left(3+4x^2\right)\sqrt{4-3x^2}}$

Let $x=\frac{1}{t}$

$dx=-\frac{1}{t^2}dt$

$=\int\frac{-\frac{1}{t^2}dt}{\left(3+\frac{4}{t^2}\right)\sqrt{4-\frac{3}{t^2}}}$

$=\int\frac{-tdt}{\left(3t^2+4\right)\sqrt{4t^2-3}}$

Now, let $4t^2-3=p^2$

$4t\,dt=p\,dp$

$=\int\frac{-\frac{p}{4}dp}{\left(\frac{3p^2+25}{4}\right)p}=-\int\frac{dp}{3p^2+25}$

$f(x)=-\frac{1}{5\sqrt{3}}\tan^{-1}\left(\frac{\sqrt{3}p}{5}\right)+C$

$f(x)=-\frac{1}{5\sqrt{3}}\tan^{-1}\left(\frac{\sqrt{3}\cdot\sqrt{4t^2-3}}{5}\right)+C$

$=-\frac{1}{5\sqrt{3}}\tan^{-1}\left(\frac{\sqrt{3}\cdot\sqrt{\frac{4}{x^2}-3}}{5}\right)+C$

$f(x)=-\frac{1}{5\sqrt{3}}\tan^{-1}\left(\frac{\sqrt{3}\cdot\sqrt{4-3x^2}}{5|x|}\right)+C$

$x=0, y=0$

$$0 = -\frac{1}{5\sqrt{3}}\tan^{-1}(\infty) + C$$

$$C = \frac{1}{5\sqrt{3}} \times \frac{\pi}{2}$$

$$f(x) = -\frac{1}{5\sqrt{3}}\tan^{-1}\frac{\sqrt{3}\sqrt{4-3x^2}}{5x} + \frac{\pi}{2\left(5\sqrt{3}\right)}$$

$$f(1) = -\frac{1}{5\sqrt{3}}\tan^{-1}\frac{\sqrt{3}}{5} + \frac{\pi}{2\left(5\sqrt{3}\right)}$$

$$= \frac{1}{5\sqrt{3}}\left(\frac{\pi}{2} - \tan^{-1}\frac{\sqrt{3}}{5}\right)$$

$$= \frac{1}{5\sqrt{3}}\cot^{-1}\frac{\sqrt{3}}{5}$$

$$= \frac{1}{5\sqrt{3}}\tan^{-1}\frac{5}{\sqrt{3}}$$

Now, $\alpha = 5,\ \beta = \sqrt{3}$

$\alpha^2 + \beta^2 = 25 + 3 = 28$